ARCore on Galaxy S9
Samsung Galaxy S9 and S9 Plus are expected to get support for ARCore in the coming weeks. **http://dgit.in/ARCoreS9**

Smartwatches aren't dead yet
According to IDC, the overall wearable market is expected to grow 13.4% by 2022, with two fifths of that comprising of smartwatches. **http://dgit.in/alivesw**

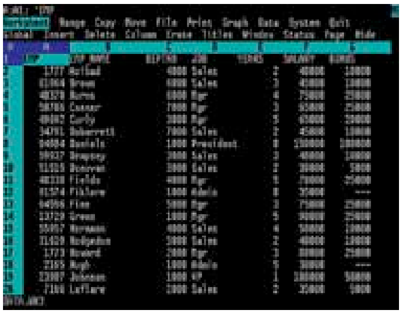
it fit within 20k of machine memory, making it suitable enough to run on a microcomputer. VisiCalc went on to become the piece of software that practically sold the Apple II by itself. The idea of marketing it for the Apple II was from Daniel Fylstra, another MIT and Harvard Business School graduate, which turned out to be a great idea as the software was to be exclusive on the Apple II for a year. During that period, quite a lot of users bought the $2000 Apple II just to use the $100 software. Unfortunately, lawsuits between Fylstra's own firm Personal Software (later renamed VisiCorp) and Software Arts distracted the developers of VisiCalc and created room for other players to take over the market.

What VisiCalc looked like

BEYOND THE PIONEERS

A former head of development at VisiCorp, Mitch Kapor founded Lotus Corporation in 1982 and released Lotus 1-2-3 in 1983, which he developed in conjunction with co-founder Jonathan Sachs. There were quite a few features that made Lotus 1-2-3 the industry standard in no time, like integrated charting, plotting and database capabilities. For instance, Lotus 1-2-3 introduced cell naming, spreadsheet macros and cell ranges. The success of this platform is evident in the fact that Lotus Development acquired Software Arts in 1985 and discontinued VisiCalc altogether.

Around the same time, Microsoft was trying to perfect its own spreadsheet software, launched in 1982 as Multiplan for the Macintosh. While it did fairly well on CP/M system, which was the mass-market operating system for Intel microcomputers at the time, on its own MS-DOS system, it lost to



Lotus 1-2-3 in 1983

the more popular Lotus 1-2-3. To turn things around, Microsoft developed a whole new software, which is what came to be known as Excel, for the Mac in 1985. A Windows version was also launched in 1987. The overall intention of Excel was to do everything that Lotus 1-2-3 does, but better. And by 1989, with Excel 3.0, Microsoft was doing exactly that. The command line interfaces of the previous competitors were no match for Excel's GUI, mouse support, 3D charts and more.

A couple of things make Excel stand out. For instance, it was among the first to offer the R1C1 addressing system along with the more familiar A1 system that we use now. Also, as its growing popularity caught the attention of the industry, it faced a lawsuit from another company that offered a software package in the financial industry with the same name. As a result, Microsoft has since been required to use 'Microsoft Excel' or 'MS Excel' for all official purposes, although that practice has dropped over time.

Excel became one of the first spreadsheets to allow users to alter the appearance of the cells itself - fonts, character attributes etc. For a more intrinsic change, it was also one of the first to offer intelligence cell recomputation - which meant that only cells directly dependent on other cells would be recalculated when a change was made.

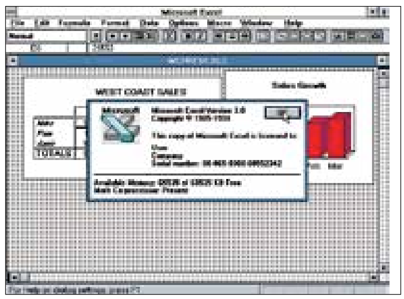
THE OFFICE AS WE KNOW IT

When it was eventually packaged with Microsoft Word and Powerpoint in the beginning of the 90s as Microsoft Office, it was the latter two who had to go through changes in design to be more consistent with Excel - so it is pretty evident who was calling the shots. By the time rival Lotus Corporation could come up with useful Windows software, Microsoft had already established its dominance with Microsoft Office. By 1995, Microsoft had successfully edged out Lotus to emerge as the top Office Productivity software provider. Eventually, IBM discontinued Lotus altogether in 2013.

Another factor that made Excel's dominance possible was the inclusion of VBA, or Visual Basic for Applications, a language based on Visual Basic which provided the ability to automate tasks in Excel to a great extent and allowed the usage of User Defined Functions (UDF). It also made spreadsheets a target for macro-viruses, which were later curbed as antivirus software developed the ability to discover them and Microsoft allowed users to disable macros altogether.

THE WEBSHEET

As with most types of software since the advent and popularity of cloud computing, spreadsheets have also been successfully implemented as



Anyone could tell what made Excel popular in an era of command line based spreadsheet tools

entirely web-based tools. These tools offer all the capabilities of their offline counterparts while adding features like multi-user collaboration, real time updates from online data-sources and more - making them true powerhouses in the office environment. As we said at the beginning of this article, without the availability of spreadsheets as they are today, this magazine would be significantly different - but then, technology always has an answer, doesn't it?